

Why Are Nigerian Children Anemic?

A closer look at which children face the highest risk, and why — based on the 2018 NDHS dataset

IHIFIX Data Analytics and AI Course — Capstone Project

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This report answers one question for a health NGO or government planner: **which children in Nigeria are most likely to be anemic, and what household factors put them at risk?** The answer matters because anemia in early childhood can slow brain development, weaken the immune system, and increase the risk of death — and because knowing who is at highest risk lets health programmes target their limited resources where they will help the most children.

Data Source

This analysis uses the **2018 Nigeria Demographic and Health Survey (NDHS)**, a nationally representative household survey covering all 36 states plus the FCT. The dataset used here includes **10,182 children under age 5** who had a hemoglobin (blood) test — the standard way anemia is measured — along with information about their household and mother.

The Headline Finding

Nearly 7 in 10 Nigerian children under 5 (68.8%) are anemic. That's the starting point, but not the interesting part — the real story is that this risk is not evenly spread. A child's chances change dramatically depending on how wealthy the household is, how much schooling the mother had, and whether the family lives in a city or village. So while the survey confirms anemia is a national problem, the data underneath it points to something more useful: a specific, identifiable group of households where the risk is concentrated — which is where a health programme's money would go furthest.

68.8%

of children under 5 are
anemic (any severity)

34-pt

gap between highest- and
lowest-risk children

3.2%

of children have SEVERE
anemia — a medical
emergency level

What This Report Covers

- **Page 2 — Background & Method:** who this covers, what data was used, why anemia matters, and how the analysis was done.
- **Pages 3–4 — Key Findings:** anemia rates broken down by wealth, education, location, health-service access, and the combined effect of all three.
- **Page 5 — Conclusion & Recommendations:** the one big takeaway and what health planners should do about it.

Background & How This Was Done

The Population and the Data

This analysis covers children aged 0–59 months and their mothers (15–49 years), drawn from the **2018 Nigeria Demographic and Health Survey (NDHS)** — a nationally representative household survey covering all 36 states plus the FCT. The dataset holds two separate anemia/hemoglobin measurements — one for the **mother** and one for the **child** (the child's version isn't adjusted for smoking, since children don't smoke, which is how the two were told apart). Only the **child's anemia level** was used, and only the 10,182 records with a valid child test were kept.

Factors analysed against it: household wealth, mother's education, urban/rural residence, mother's age group, bed net access, recent fever, and iron supplement use.

Why Anemia Matters

Anemia means a child's blood doesn't carry enough oxygen — usually from poor iron intake, malaria, frequent illness, or poor sanitation. Anemic children tire more easily, get sick more often, and if untreated can suffer permanent effects on brain development and learning. It's one of the core child-health indicators tracked by WHO and UNICEF, which is part of why it's worth analysing carefully rather than just reporting the national average.

Research Question

Which socioeconomic and household factors are most strongly linked to anemia among Nigerian children under 5, and how large is the combined effect of these factors together? This was chosen over a simpler question because the answer is directly actionable — it tells a health programme which households to prioritise, rather than just restating that anemia is common.

Method, Cleaning & Tools

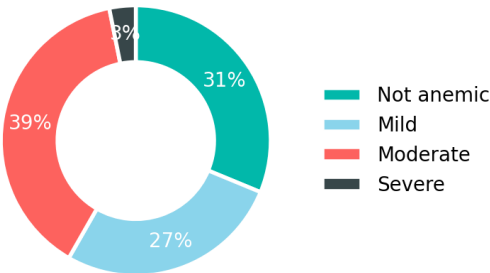
Records with no child test result were excluded, and “Mild/Moderate/Severe” were grouped into one “Anemic” category for comparison, alongside looking at severity on its own. The data was cleaned and queried in **Microsoft SQL Server**, cross-checked with **Excel** pivot tables, and the final charts were built in **Power BI**. The main technique is cross-tabulation with grouped averages — kept deliberately simple, since the gaps between groups turned out to be large and consistent without needing a full statistical model.

Key Findings — Part 1

Each chart below compares the share of children who are anemic across a household factor.

Overall Anemia Burden

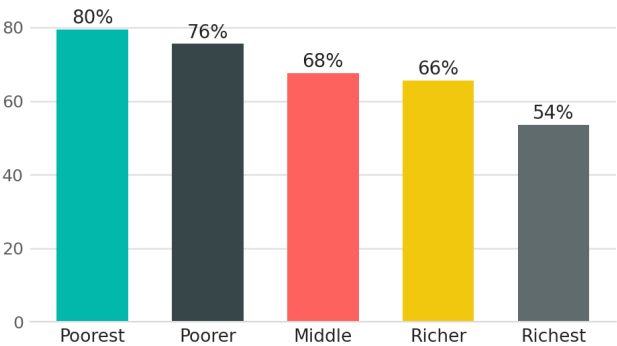
Anemia Severity Among Children Under 5



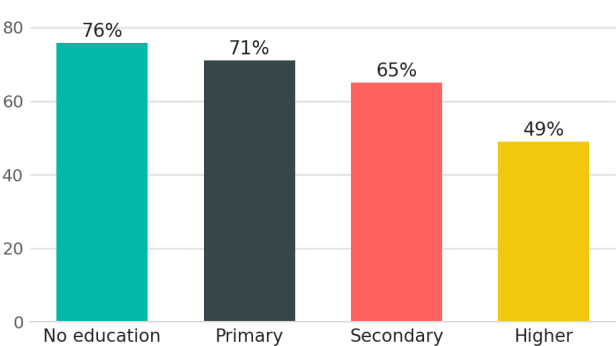
Key insight: 68.8% of children under 5 tested anemic. Most of this is Mild or Moderate anemia (65.6% combined), but 3.2% of ALL children — about 1 in 31 — have Severe anemia, a level that typically requires urgent medical treatment.

Anemia by Wealth and by Mother's Education

By Household Wealth



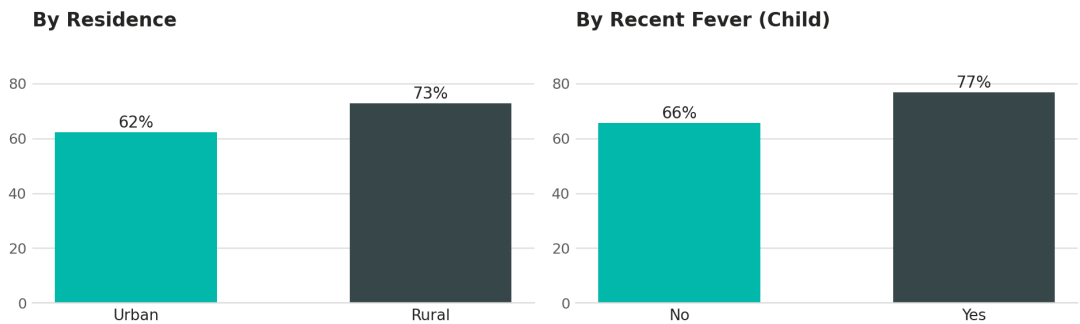
By Mother's Education



Key insight: anemia falls steadily as wealth rises — from 79.5% in the poorest households to 53.5% in the richest, a 26-point gap. It falls just as steadily with education: 75.8% for children of mothers with no schooling versus 49.0% for children of mothers with higher education — a 27-point gap. Wealth affects diet, treatment speed, and clean water; education affects how early illness is recognised and acted on.

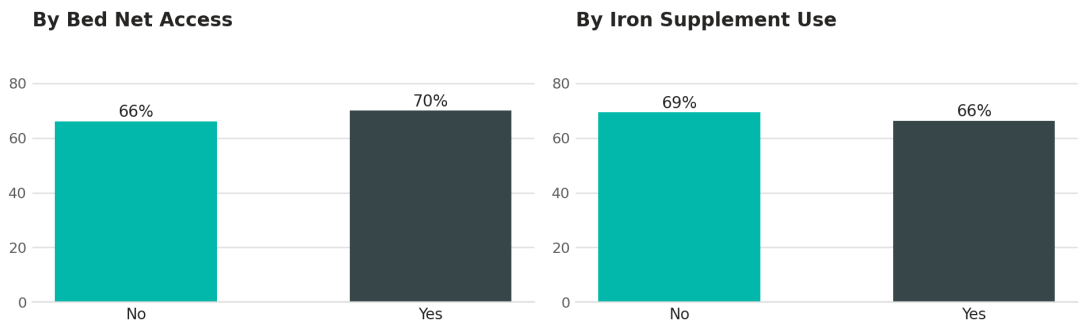
Key Findings — Part 2

Location and Recent Illness



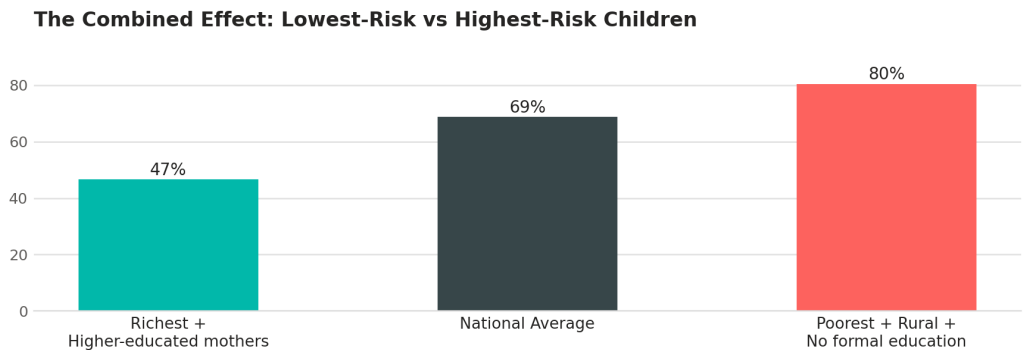
Key insight: rural children are anemic more often than urban children (72.9% vs 62.3%). Children who had a fever in the last two weeks are anemic more often too (76.9% vs 65.7%), confirming that infection is a major driver of childhood anemia in Nigeria, not just diet.

Bed Nets and Iron Supplements — a Caution



Key insight: children with a bed net at home, and children taking iron supplements, show slightly HIGHER anemia rates, not lower — very unlikely to mean these interventions cause anemia. It far more likely reflects targeting: bed nets and supplements are typically given where risk is already highest, so this simple comparison mixes cause and effect. It should not be read as “bed nets don’t work.”

The Combined Effect



Key insight: when the three strongest factors are stacked together, the gap becomes extreme: children who are poor, rural, and whose mother has no formal education are anemic 80.4% of the time, versus 46.6% for children from the richest, higher-educated households — a 34-point gap, nearly double the risk. This is arguably the most useful chart in the report for deciding where to target resources.

Conclusion & Recommendations

The Big Takeaway

Childhood anemia in Nigeria is not random — it is heavily concentrated in poor, rural households where mothers have little or no formal education. A child born into this combination faces roughly **double the anemia risk** of a child born into a wealthy, educated, urban household (80.4% vs 46.6%). Because wealth, education and location tend to overlap in the same families, anemia is not a scattered problem to fix household-by-household — it is a **concentrated problem** that can be tackled by targeting a specific, identifiable group of communities.

Recommendations

- **Target the combination, not just one factor.** Programmes that only look at “poor households” or only “rural areas” will miss the sharpest risk group. Prioritise rural communities with high poverty AND low female education together — the data shows this is where risk compounds.
- **Treat fever and anemia together.** Because recent fever is strongly linked to anemia, malaria prevention and treatment services should be paired with iron/anemia screening at the same clinic visit, not run as separate programmes.
- **Invest in maternal education and health literacy.** The education gap (27 points) is nearly as large as the wealth gap, suggesting that even modest improvements in mothers' health knowledge — not just household income — could meaningfully reduce child anemia.
- **Re-audit bed net and iron supplement targeting.** The fact that current recipients of bed nets and iron supplements still show high anemia rates suggests these programmes are (correctly) reaching high-risk families, but may need to be paired with follow-up and dosage/compliance checks to actually lower anemia in those households over time.
- **Flag the 3.2% with severe anemia for urgent follow-up.** This group needs clinical treatment, not just prevention — they should be identifiable and traceable through routine health visits.

A Note on What This Data Can and Can't Tell Us

This analysis shows strong, consistent patterns — it does not prove that any single factor directly causes anemia on its own. Wealth, education, location, and health access are all tangled together in real life. The bed net and iron-supplement result above is a reminder that a simple comparison can go backwards when the group being compared was chosen because it was already high-risk. For a next phase, a statistical model that controls for multiple factors at once would help confirm which factor matters most on its own.